

Scraftie — Personal Music Streaming Server

SERVER PERMISSIONS & SHARED GROUP CONFIGURATION

1. Problem Overview & Root Cause Analysis

Initial Permission Structure

The music library on the server was structured with the following directory ownerships:

- `/music` — Owner: `<user_account>`, Group: `<user_group>`
- `/music/OG_SD_Card` — Owner: `navidrome`, Group: `navidrome`
- `/music/OG_SD_Card/<Artist>` — Owner: `navidrome`, Group: `navidrome`

The Issue & Verification

When attempting to write to the library using the primary Linux user account (`<user_account>`):

```
touch /music/OG_SD_Card/<Artist>/test.txt
```

The system denied access with the verified error:

```
Permission denied
```

This strict permission model prevented remote uploads over SCP when logged in under the standard user account.

Root Cause Analysis

Directories under `/music/OG_SD_Card` were exclusively owned by the service user and group `navidrome:navidrome`. Because the primary user account was neither the file owner nor a member of the `navidrome` group, standard POSIX filesystem permissions blocked all write attempts.

2. Architecture & Design Decisions

Design Choice: Shared Group vs. Full Ownership Change

Rather than transferring complete ownership of `/music/OG_SD_Card` to the primary user account or opening world-write permissions (`777`), a dedicated shared group strategy was implemented.

- **Service Continuity:** Navidrome runs under the dedicated `navidrome` service account and requires consistent read/write access for scanning and indexing media files.
- **Principle of Least Privilege:** Creating a shared Linux group (`music`) allows both human and service accounts equal management access without compromising system security or altering service user identities.

3. Implementation & Commands Executed

Step 1: Shared Group Creation

A shared Linux group named `music` was created, and both users were assigned to it:

- Group name: `music`
- Members: `<user_account>` `navidrome`

4. Access Control Lists (ACL) & Verification

ACL Configuration Executed

Verification confirmed that ACL utility tools were installed on the host OS (`getfacl 2.3.2`). Explicit group and default inherited ACL rules were configured:

```
sudo setfacl -m g:music:rwx /music/OG_SD_Card
sudo setfacl -d -m u::rwx,g::rwx,o::rx /music/OG_SD_Card
```

Verified ACL Output

The applied permissions on `/music/OG_SD_Card` were verified via `getfacl` :

```
# owner: navidrome
# group: music
# flags: -s-
user::rwx
group::rwx
group:music:rwx
mask::rwx
other::r-x
default:user::rwx
default:group::rwx
default:other::r-x
```

Detailed Rule Interpretation

- `flags: -s-` — Confirms active enforcement of the `setgid` bit across the directory structure.
- `group:music:rwx` — Guarantees explicit read, write, and traverse privileges for members of the `music` group.
- `default:user::rwx` & `default:group::rwx` — Inherited rule configuration. Ensures newly created files or folders automatically inherit full `rwx` rights for both the owner and the `music` group regardless of client umask configurations.
- `default:other::r-x` — Preserves read and execute access for unprivileged users while preventing unauthorized writes.

5. Album Ingestion Case Study: Sample Media Upload

Transfer Verification

The first album uploaded following group configuration was a test media folder. The transfer was executed from the local client machine over Tailscale using SCP:

```
scp -r "C:\Users\\Music\ - <Album>" <user_account>@<TAILSCALE_IP>:/music/OG_SD_Card/<Artist>/
```

The transfer completed successfully (containing 20 MP3 audio tracks and `cover.jpg` , totaling ~180 MB).

Manual Correction Analysis

Following the initial upload, directory permissions were audited:

```
drwx----- 2 <user_account> music
```

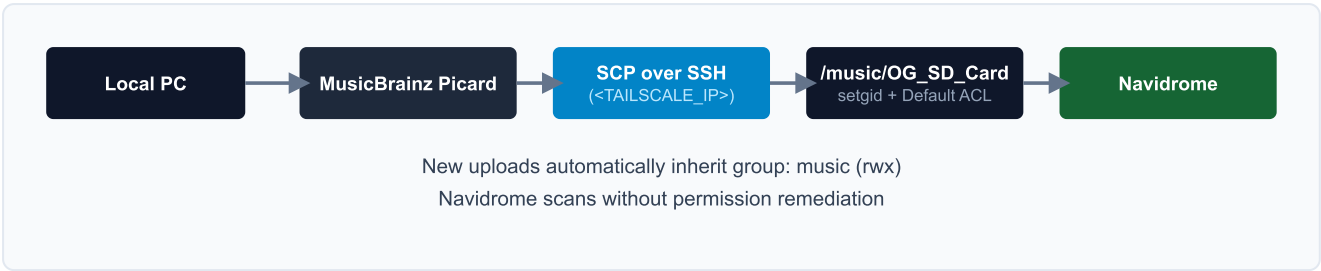
Cause: The target album directory had been created prior to applying the recursive default ACL rules across the

6. Tailscale Remote Ingestion Workflow

The Scraftie server remains reachable over Tailscale via the designated Tailscale IP (`<TAILSCALE_IP>`). SSH connectivity was verified using:

```
ssh <user_account>@<TAILSCALE_IP>
```

End-to-End Media Ingestion Pipeline



Expected Future Behavior

- **Automated Inheritance:** All future music uploads via SCP will automatically inherit group ownership of `music` due to the active `setgid` bit.
- **ACL Enforcement:** The default ACL rule ensures new items receive full group `rwx` permissions immediately upon creation.
- **Zero Maintenance Scanning:** Navidrome retains uninterrupted read/write access to index newly added albums without requiring manual `chmod` intervention.

7. Summary Audit & Verification Log

Operation / Test	Executed Command	Verified Result Status
Initial Write Test	<code>touch /music/OG_SD_Card/<Artist>/test.txt</code>	PERMISSION DENIED (Verified)
Group Ownership	<code>sudo chown -R navidrome:music /music/OG_SD_Card</code>	SUCCESS Group set to music
Group Permissions	<code>sudo chmod -R g+rwX /music/OG_SD_Card</code>	SUCCESS Modes updated
Setgid Bit Setup	<code>sudo find /music/OG_SD_Card -type d -exec chmod g+s {} \;</code>	SUCCESS Bit <code>g+s</code> set on dirs
ACL Tooling Check	Binary verification	VERIFIED <code>getfacl 2.3.2</code>
Default ACL Setup	<code>sudo setfacl -d -m u::rwx,g::rwx,o::rx /music/OG_SD_Card</code>	SUCCESS Rules set
ACL Audit	<code>getfacl /music/OG_SD_Card</code>	VERIFIED Output contains <code>flags: -s-</code> & defaults
SSH Verification	<code>ssh <user_account>@<TAILSCALE_IP></code>	VERIFIED Remote login successful
Album SCP Transfer	<code>scp -r ... <user_account>@<TAILSCALE_IP>:...</code>	SUCCESS Media folder (~180 MB)
Album Correction	<code>chmod 770 "/music/OG_SD_Card/<Artist>/<Artist> -<Album>"</code>	VERIFIED Mode updated to <code>drwxrwx---</code>
File Lifecycle Test	Created & deleted test file as primary user in target folder	VERIFIED Create & delete successful